

Exercise 1

Consider a planar 3R robot with equal links of length $L > 0$ that is commanded with joint velocities $\dot{q} \in \mathbb{R}^3$. Each joint has the same available motion range

$$q_i \in [q_{i,min}, q_{i,max}] = [-\Delta, +\Delta], \quad \text{with } \Delta = \frac{3\pi}{4} [\text{rad}], \quad \text{for } i = 1, 2, 3.$$

The joint range function is defined in general as

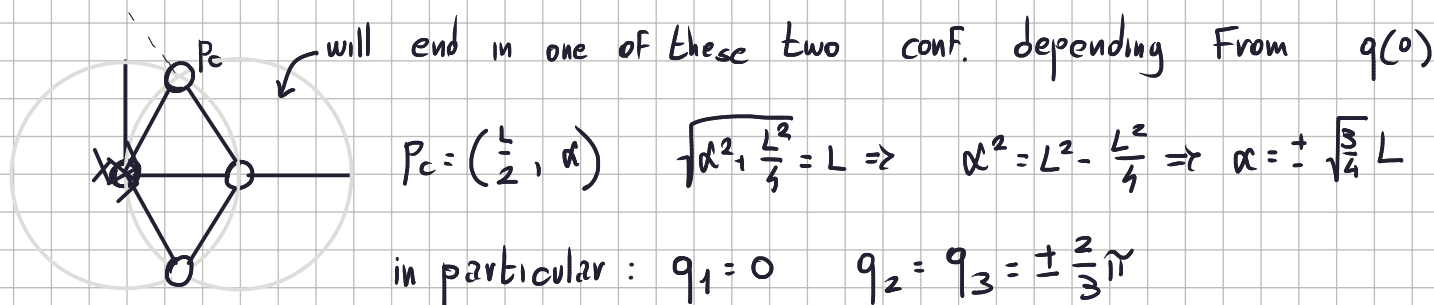
$$H(q) = \frac{1}{2N} \sum_{i=1}^N \left(\frac{q_i - \bar{q}_i}{q_{i,max} - q_{i,min}} \right)^2, \quad \text{with } \bar{q}_i = \frac{q_{i,min} + q_{i,max}}{2},$$

$\frac{\partial H}{\partial q_i} = \frac{2}{3\pi N} q_i = \frac{2\pi}{9} q_i$ so $\nabla H = \frac{2\pi}{9} (q_1 \ q_2 \ q_3)^T$ hence $-\nabla H$ always points to $(0 \ 0 \ 0)^T$. Hence $\dot{q} = -(\mathbf{I} - \mathbf{J}^* \mathbf{J}) \nabla H$ where

$$\mathbf{J} = L \begin{pmatrix} -s_1 - s_{12} - s_{123} & -s_{12} - s_{123} & -s_{123} \\ c_1 + c_{12} + c_{123} & c_{12} + c_{123} & c_{123} \end{pmatrix}. \quad H=0 \Leftrightarrow q=0 \Rightarrow p \neq 0 \quad \text{hence the final conf.}$$

won't be $q=0 \Rightarrow H > 0$. This happens when ∇H is orthogonal to the plane $\mathbf{J}\dot{q}=0$. Obs 1: When $p=0$, any $\dot{q}_1 \neq 0$ won't change the final conf, hence in the final conf $q_1=0$.

The third joint must be at distance L from the origin and the second is in $(L, 0)$ hence we are in the following situation.



Exercise 2

$$T_1 = \frac{1}{2} \mathbf{I}_1 \dot{q}_1^2 \quad T_2 = \frac{1}{2} \mathbf{I}_2 \dot{q}_1^2 + \frac{1}{2} m_2 (q_2 - d_2)^2 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2$$

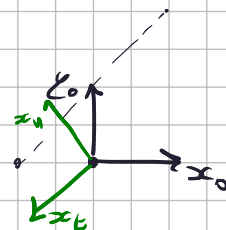
$$M_{11} = \mathbf{I}_1 + \mathbf{I}_2 + m_2 q_2^2 + m_2 d_2^2 - 2m_2 d_2 q_2 \quad M_{22} = m_2$$

$$\mathbf{I}_1 + \mathbf{I}_2 + m_2 d_2^2 = c_1 \quad m_2 = c_3 \quad -2m_2 d_2 = c_2$$

$$\mathbf{M} = \begin{pmatrix} c_1 + c_2 q_2 + c_3 q_2^2 & 0 \\ 0 & c_3 \end{pmatrix}$$

$$C_1 = \left(\frac{1}{2} c_2 + c_3 q_2 \right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow C_1 = (c_2 + 2c_3 q_2) \dot{q}_1 \dot{q}_2$$

$$C_2 = -\left(\frac{1}{2} c_2 + c_3 q_2 \right) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow C_2 = -\left(\frac{1}{2} c_2 + c_3 q_2 \right) \dot{q}_1^2$$



The tangent direction of the path is ${}^0 x_t = \frac{1}{5} (4 \ 3)$ the normal is ${}^0 x_n = \frac{1}{5} (-3 \ 4)$ the transf. between RF_0 is a rotation along z of $\theta = \tan^{-1} \frac{3}{4} \Rightarrow {}^0 R_{Ln} = \begin{pmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{pmatrix}$.

Now we want to define the DK in RF_{Ln}

$${}^0P = \begin{Bmatrix} q_2 C_1 \\ q_2 S_1 \end{Bmatrix} \Rightarrow {}^{Ln}P = {}^0R_{Ln}^T {}^0P = \begin{pmatrix} 4/5 & 3/5 \\ -3/5 & 4/5 \end{pmatrix} \begin{Bmatrix} q_2 C_1 \\ q_2 S_1 \end{Bmatrix} = \begin{Bmatrix} \frac{q_2}{5}(4C_1 + 3S_1) \\ \frac{q_2}{5}(4S_1 - 3C_1) \end{Bmatrix} \quad \text{hence}$$

$$J = \begin{pmatrix} \frac{q_2}{5}(3C_1 - 4S_1) & \frac{1}{5}(4C_1 + 3S_1) \\ \frac{q_2}{5}(4C_1 + 3S_1) & \frac{1}{5}(4S_1 - 3C_1) \end{pmatrix} \quad \text{hence we express the dynamic model in } RF_{Ln}$$

$$M_\chi = J^T M J^{-1} \quad c_\chi = J^T (c - M \dot{J} J^{-1} \dot{\chi}) \Rightarrow M_\chi \ddot{\chi} + c_\chi = \overset{\substack{\uparrow \\ \text{control} \\ \text{force}}}{F}$$

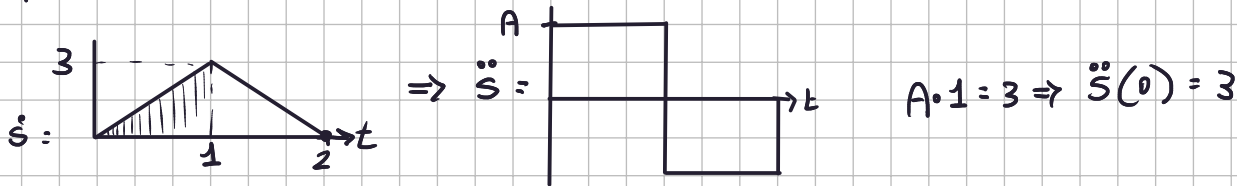
We consider $F = c_\chi + M_\chi a \Rightarrow \ddot{\chi} = a$ and then

$a = \ddot{p}_d(t) + K_p e + K_d \dot{e}$ where e and $\dot{e}$ are the position and velocity error in the task space in RF_{Ln} hence

$$K_p = \begin{pmatrix} 4 & 0 \\ 0 & 16 \end{pmatrix} \quad K_d = \begin{pmatrix} 4 & 0 \\ 0 & 8 \end{pmatrix} \Rightarrow \text{achieves the desired behaviour.}$$

Now we need the expr. of the initial acceleration to execute.

$$p_d(s) = p_i + s(p_f - p_i) \quad s \in [0, 1] \quad s(t) = b \cdot c \cdot b \quad t \in [0, 2]$$



$$\ddot{p}_d(0) = \frac{1}{5} \begin{pmatrix} 4 & 3 \end{pmatrix} \cdot \ddot{s}(0) = \frac{3}{5} \begin{pmatrix} 4 & 3 \end{pmatrix} \quad \text{Now i compute numerically all the terms for } z.$$

$$e_1 = 102.5 \quad e_3 = 10 \quad e_2 = -50$$

$$M = \begin{pmatrix} 42.5 & 0 \\ 0 & 10 \end{pmatrix} \quad J = \frac{1}{5} \begin{pmatrix} 6 & 4 \\ 8 & -3 \end{pmatrix} \quad c = c_\chi = 0 \quad M_\chi = \begin{pmatrix} 10.225 & 0.3 \\ 0.3 & 10.4 \end{pmatrix}$$

$$e = \begin{pmatrix} \frac{q_2}{5}(4C_1 + 3S_1) \\ \frac{q_2}{5}(4S_1 - 3C_1) \end{pmatrix} - {}^0R_{Ln}^T \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -9 \\ -12 \end{pmatrix} \quad \dot{e} = 0 \quad \text{since } \dot{p}_d(0) = 0$$

$$z(0) = \ddot{p}_d(0) + K_p e + K_d \dot{e} = \frac{3}{5} \begin{pmatrix} 4 & 3 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ 0 & 16 \end{pmatrix} \frac{1}{5} \begin{pmatrix} -9 \\ -12 \end{pmatrix} = -\frac{1}{5} \begin{pmatrix} 24 \\ 183 \end{pmatrix}$$

$$F = M_\chi z = \begin{Bmatrix} -60.06 \\ -382.08 \end{Bmatrix} \quad z = J^T F = \begin{pmatrix} -683.4 & 181.2 \end{pmatrix}^T$$

at $t = \frac{T}{2}$ the position will be $p_d(\frac{T}{2}) + \begin{pmatrix} e_6 \\ e_n \end{pmatrix}$ where $\begin{Bmatrix} e_6 \\ e_n \end{Bmatrix}$ is the error that is close to zero given the expressions.

e_t = no overshoot profile e_n = overdamped

Exercise 3

For the system (a): $(M+B)\ddot{\theta} + Mg_0 = F$, with the law:

$$(M+B)\ddot{\theta} + Mg_0 = K_P(\theta_d - \theta) - K_D\dot{\theta} + K_I \int (\theta_d - \theta) dt \quad \text{we move to the Laplace domain:}$$

$$(M+B)\theta s^2 + Mg_0 = K_P\theta_d - K_P\theta - K_D\theta s + \frac{1}{s}K_I\theta_d - \frac{1}{s}K_I\theta \Rightarrow$$

$$\theta \left((M+B)s^2 + K_D s + K_P + \frac{1}{s}K_I \right) = K_P\theta_d - Mg_0 + \frac{1}{s}K_I\theta_d \Rightarrow$$

$$\theta(s) = \frac{K_P\theta_d - Mg_0 + \frac{1}{s}K_I\theta_d}{(M+B)s^2 + K_D s + K_P + \frac{1}{s}K_I} = \frac{s(K_P\theta_d - Mg_0) + \theta_d K_I}{(M+B)s^3 + K_D s^2 + K_P s + K_I}$$

with a suitable choice of K_P, K_D, K_I the roots[↑] have neg. real part and

$$\lim_{t \rightarrow \infty} \theta(t) = \lim_{s \rightarrow 0} s \frac{K_P\theta_d - Mg_0 + \frac{1}{s}K_I\theta_d}{(M+B)s^2 + K_D s + K_P + \frac{1}{s}K_I} = \theta_d$$

With the iterative law: $(M+B)\ddot{\theta} + \underset{(M+B)g_0}{\overset{K_P(q_d - \theta_i) + v_{i-1}}{\ddot{\theta}}} = K_P(\theta_d - \theta) - K_D\dot{\theta} + v_i$

At the steady state of i -th iteration $v_i = Mg_0$ hence after the first

$$it: (M+B)\ddot{\theta} + Mg_0 = K_P(\theta_d - \theta) - K_D\dot{\theta} + Mg_0 \Rightarrow (M+B)\ddot{\theta} = K_P(\theta_d - \theta) - K_D\dot{\theta}$$

$\uparrow$ PD $\Rightarrow$ converge.

The system in (b) is:

$$\begin{cases} B\ddot{\theta} + K(\theta - q) = F \\ M\ddot{q} + K(q - \theta) + Mg_0 = 0 \end{cases}$$